

**The Determinant and Causes of High Unemployment Rate in
Nigeria**

Case Study: Year 2000 to 2022

Diploma Thesis

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Statutory Declaration

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Abstract

This thesis investigates the factors that contribute to Nigeria's high unemployment rates, a serious problem that has seriously impeded the nation's socioeconomic advancement. The study looks into both cyclical and structural causes of long-term unemployment, such as macroeconomic volatility, poor educational systems, restricted access to capital, and a mismatch between labor force capabilities and market demands. It also looks at how economic policies, demographic pressures—particularly the youth bulge—and political instability affect employment trends. This study identifies the main obstacles to job creation using a combination of qualitative and quantitative methods, such as data analysis, literature review, and interviews with important stakeholders. It also suggests policy recommendations meant to promote sustainable economic growth, enhance education and skill development, and create an environment that is favorable to job creation. The results demonstrate the intricacy of Nigeria's unemployment issue and the necessity of all-encompassing, multidimensional actions to deal with the underlying reasons and lower the unemployment rate.

Keywords: Unemployment, GDP growth, welfare, cyclical, poverty

Abstracktni

Factory křispívají k vysoké míře nezaměstnanosti v Nigérii, který vážně narušuje socioekonomický rozvoj země, což je vážný problém. The study's objectives are to determine the cyklickými, tak strukturálními příčinami dlouhodobé nezaměstnanosti, jako je makroekonomická volatilita, špatné vzdělávací systémy, and omezený přístup from pracovní síly to požadavky trhu. Politická nestabilita ovlivňují trendy zaměstnanosti, zabývá se také tím, jak hospodářské politiky, demografické tlaky – zejména nárůst mladých lidí. According to this study, přehled literature and rozhovory s důležitými zainteresovanými stranami překážky vytváření pracovních míst pomocí kombinace kvalitativních a kvantitativních metod. In navrhuje, která mají podporovat udržitelný hospodářský růst, zlepšit vzdělávání a rozvoj proveností a vytvořit prostředí příznivé pro vytváření pracovních míst. výcerozměrných opatření k řešení základních důvodů a snížení míry nezaměstnanosti v Nigérii a nezbytnost všezahrnujících problému nezaměstnanosti v Nigérii.

Klíčová slova: Nezaměstnanost, růst HDP, blahobyt, cyklický, chudoba

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1. INTRODUCTION

1.1 Background and Context

The main objective of macroeconomic in every economy is to focus on full employment, price stability, and rapid production growth. These indicators are independent thus, changes to one of these inevitably affect the others. For instance, during a recession, prices decline, unemployment increases, and output declines. Therefore, it's critical to employ policy instruments sparingly in order to prevent unexpected consequences. One of the main macroeconomic issues that most emerging nations deal with is unemployment. People that are eager and able to work are in this circumstance where they are unable to obtain jobs that will allow them to support themselves. Nigeria has a protracted history of high rates of unemployment. This has had numerous detrimental effects on the nation. It has caused the output and income growth rates to continuously decline, which has resulted in a low standard of life.

According to Alanana (2003), unemployment is potentially dangerous as it sends disturbing signal to all segments of the Nigerian society. Nigeria is currently experiencing a recession with a 33% unemployment rate. The fact that thousands of college grads are doing nothing to make ends meet makes this quite unsettling. Robberies, drug trafficking, and kidnappings have all grown as a result of this. Additionally, it has exacerbated the nation's development issues by increasing the flow of human capital to industrialized nations.

Unemployment, as a global socioeconomic challenge, poses significant impediments to individual livelihoods and national development. Given that Nigeria is the most populated country in Africa and a major contributor to the continent's economy, it is critical to comprehend the complex factors that contribute to unemployment. The challenges of unemployment extend beyond economic indicators, impacting social cohesion, public health, and overall well-being. As Nigeria grapples with this multifaceted issue, there is an urgent need to dissect the complex web of factors that contribute to the persistent high levels of unemployment.

Nigeria, endowed with abundant natural resources and a diverse population, has experienced periods of economic growth and contraction. However, the shadow of unemployment persists, casting doubt on the realization of the demographic dividend and economic potential. The country's dependence on oil, while a significant revenue source, has led to structural imbalances

and hindered the development of other sectors (Ajakaiye & Jerome, 2000). As Nigeria strives for economic diversification and inclusive growth, the complexities of unemployment warrant a comprehensive examination.

1.2 Global Perspectives on Unemployment

A foundational understanding of unemployment draws on global perspectives that recognize both cyclical and structural factors as contributors (Blanchard & Summers, 1986). The cyclical nature relates to fluctuations in economic cycles, while structural factors encompass long-term issues such as education, technology, and government policies. Applying this global framework to Nigeria allows for a nuanced exploration of the interplay between international economic trends and the nation's internal dynamics.

1.3 Historical Overview of Unemployment in Nigeria

A historical lens reveals the evolution of unemployment in Nigeria, particularly in the wake of significant economic shifts. The discovery of oil in the mid-20th century marked a turning point, leading to a neglect of other sectors and contributing to the country's vulnerability to oil price fluctuations (Ajakaiye & Jerome, 2000). Understanding this historical trajectory is crucial for contextualizing the contemporary challenges and formulating effective strategies.

1.4 Conceptual Framework

Within a specific field of study, a conceptual framework is a theoretical model or structure that helps to organize and make sense of complicated facts or concepts. It offers a structure for deciphering, evaluating, and elucidating connections between different variables or components.

Conceptual

Conceptual frameworks are used as guiding frameworks for research, theory development, and practical applications in a variety of fields, including education, psychology, economics, and social sciences.

1.4.1 Structural Determinants

Education and skill gaps are recognized as crucial structural determinants of unemployment (Becker, 1993). Nigeria's education system, although making strides, grapples with challenges such as outdated curricula and a mismatch between skills acquired and market demands (World Bank, 2018). An exploration of these structural issues is essential for aligning the workforce with the dynamic needs of a modern and diverse economy.

1.3.2 Policy Interventions and Socio-Economic Dynamics

Government policies play a central role in shaping employment dynamics. Initiatives like the National Economic Empowerment and Development Strategy (NEEDS) underscore the government's commitment to fostering economic growth and job creation (Federal Ministry of Budget and National Planning, 2016). Evaluating the effectiveness of these interventions is crucial for refining strategies to address unemployment. Additionally, an examination of socio-economic dynamics, including gender disparities and regional variations, provides a holistic understanding of the challenges faced by different segments of the population.

1.4 Objectives of the Study

This research aims to achieve the following objectives:

- Identify and analyse structural challenges contributing to unemployment in Nigeria.
- Assess the effectiveness of existing policies in addressing unemployment.
- Explore socio-economic dynamics, including gender and regional variations, as determinants of unemployment.
- Provide insights for evidence-based policy recommendations to mitigate unemployment in Nigeria.

1.5 Significance of the Study

In Nigeria, unemployment is one of the most urgent socioeconomic problems. The nation has one of the highest rates of unemployment in the world, and the rate for young people is particularly concerning. The difficulties of unemployment are frequently highlighted in World Bank and Nigerian government publications, especially for recent graduates who struggle to obtain employment in the formal sector. One of the main causes of poverty, inequality, and social instability in the nation as of late has been shown to be unemployment. Particularly among young people, unemployment is a serious issue since it can lead to social discontent and raise the likelihood of criminal activity. Its capacity to investigate the extent of unemployment in Nigeria, comprehend its root causes, and provide suggestions for resolving this issue at both the macro and local levels makes this initiative significant.

In-depth analysis of the unemployment rate, its historical patterns, and its effects on different demographic groups—particularly with regard to age, education, and location—are provided by the project. The study will illustrate the differences between urban and rural locations, as well as

the variances among Nigerian states, by examining the unemployment rate. This is significant since the effects of unemployment vary depending on a number of circumstances, including the availability of job prospects, educational chances, and the level of economic growth in a given area. For instance, a lack of infrastructure and economic activity causes increased unemployment in some states, especially in the northern region, whereas other states with robust industrial and agricultural sectors might offer more job prospects. This effort contributes to the mapping of these discrepancies and offers a sophisticated understanding of the regional variations in unemployment. Examining the factors that contribute to unemployment in Nigeria is equally important. There are several factors that contribute to unemployment, including social, political, and economic factors. One of the main causes of the study will be the misalignment between the educational system and the labor market. A considerable underemployment rate and a growing number of graduates who are unable to find employment that matches their qualifications are caused by Nigeria's educational system, which regularly produces graduates who lack the skills needed by the industry. This mismatch between educational production and labor demand is one of the primary reasons of youth unemployment in particular.

Nigeria's economic situation is another important factor. Significant economic difficulties have beset Nigeria in recent decades, including recessions, volatile oil prices, and a lack of economic diversification. Oil exports continue to be a major part of Nigeria's economy, and the availability of jobs in industries like manufacturing and services has been directly impacted by fluctuations in oil prices worldwide. The demand for work in these areas is further exacerbated by the expanding population, and the lack of economic diversification means that prospects outside the oil sector are restricted.

Nigerian unemployment is mostly caused by political unrest and problems with administration in addition to economic concerns. The main obstacles to successful job creation are political will to undertake job development programs, public sector inefficiencies, and corruption. The study will investigate how these elements lead to a shortage of job prospects, in addition to poor policy execution and insufficient infrastructure. The study will also examine the role of government programs and initiatives, like the Social Investment Programmes (SIPs) and the National Directorate of Employment (NDE), that are designed to reduce unemployment. Ultimately, this study is important because it has the ability to influence public policy and offer insights that can direct initiatives to address unemployment in Nigeria. Through an analysis of the

unemployment rate and its drivers, the project will provide suggestions for policy changes that target the underlying causes of unemployment. These could include boosting economic diversity, improving infrastructure and governance, bolstering social safety nets for the unemployed, and revamping the educational system to better meet the demands of the labor market. Additionally, the project's results will add to the global conversation on unemployment in developing nations by providing a case study on how populous nations and emerging economies may more effectively handle the difficulties associated with economic growth and job creation. To sum up, this research is essential for understanding Nigerian unemployment as well as for preparing for a time when all Nigerians will have access to long-term job opportunities.

1.6 Structure of the Thesis

This thesis is organized into distinct chapters, each focusing on a specific aspect of the determinants and causes of unemployment in Nigeria. Chapter 2 delves into a detailed literature review, providing a comprehensive synthesis of existing research. Subsequent chapters will explore structural determinants, policy interventions, socio-economic dynamics, and offer recommendations for addressing unemployment in Nigeria.

In conclusion, the exploration of unemployment in Nigeria is not merely an academic endeavour but a critical pursuit for fostering sustainable development and enhancing the quality of life for its citizens. This research endeavours to contribute to this vital discourse by providing a nuanced understanding of the determinants and causes of unemployment, paving the way for informed and effective interventions.

1.7. Theoretical considerations

Since both GDP and unemployment are rising over the long term, Walterskirchen (1999) argues that there can be no negative relationship between economic growth and unemployment. Since labor-market reforms will lower wage costs and increase employment, Njoku and Ihugba (2011) assert that they will unavoidably lead to higher output throughout the adjustment process.

Comparably, it is also possible to make the case that increased employment of low-skilled people promotes growth because it is more advantageous to invest in human capital when combined with a larger pool of low-skilled workers. Higher employment also implies higher accumulation of human capital if job training is the primary cause of increasing employment (Aghion and Howitt,

1994; Davcri and Tabellini, 2000). The average length of time children spend in school before entering the workforce, on the other hand, is likely to decrease as employment rises (Storesletten and Zilibotti, 2000). From 6.1 in 1985 to 19.7 in 2009, Nigeria's unemployment rate increased by 69%; in 2010 and 2011, it increased to 21.1 and 23.9 percent, respectively. The economy grew by 72% between 1995 and 2011, from 281,407.4 to 814,407.0. This suggests that unemployment and economic growth were occurring at the same pace. Why? The regularity of Unemployment rises in tandem with the nation's population growth. A larger labor pool is indicated by an increase in the population. With a population growth of (88,992,220-140,003,542) 36.4 percent and an economic growth of 55.5 percent over a 16-year period (1991-2006), unemployment should have dropped significantly but instead rose by 74.8 percent. In contrast, the oil industry, which employs less than 10 percent of the labor force, contributed 26.7 percent of the nation's GDP on average between 1995 and 2011, while agriculture, which accounts for the majority of gainful employment in the country, contributed 39.3 percent, only 12.6 percent less than the oil industry. More over half of the country's workforce is employed in agriculture, which grew its share of GDP from 34.1 percent in 1995 to 43.5 percent in 2011. This amounts to an increase of 9.4%. The unemployment rate rose by 69% in spite of this gain (CBN, 2012). This is related to the important but underappreciated sectors, such as agriculture, where the country has a competitive edge. Inadequate infrastructure in the country, which impeded rapid industrialization; blatant mismanagement of national resources, which led to wasteful spending and the misappropriation of funds and priorities; and a lack of political will to implement certain policies that could open up the Nigerian economy to the private sector engagement are some of the other factors contributing to this unusual state of affairs. Among these is the government's hesitation to fully de-regulate the downstream oil and gas sector. Bello (2003) asserts that the insufficient educational system, which fails to equip youth with the requisite technical and entrepreneurial skills, is another contributing cause.

2. LITERATURE REVIEW

2.1 Overview of unemployment in Nigeria

A politician named Calvin Coolidge claims that unemployment occurs "when a great number of individuals can't look for a decent job." Nigeria's unemployment rate increased from 23.1% in the third quarter of 2018 to 27.1% in the second quarter of 2020, a 4% increase. According to a National Bureau Statistics report, Nigeria's underemployment rate increased from 20.1% in 2018 to 28.6%. If an individual participates in the production of goods and services, so contributing to the GDP, a component of the national accounts, and receives compensation in whatever form or amount for that activity, that individual is deemed to be employed. Goods (GDP) should be produced domestically in Nigeria in order to promote employment and reduce unemployment. The term "unemployed" refers to somebody who is able to work but does not currently hold employment. One of the most pervasive and enduring problems in the world is unemployment. People as well as international communities are in question. A worker's percentage of the total available workforce that is jobless but actively looking for work is used to calculate unemployment. This is referred to as the unemployment rate. Nevertheless, it is evident that the rate of depressing unemployment has accelerated in recent years as a result of the economic downturn (Olawale 2016). The difference between the number of people who are currently employed and the number of people who could be employed is another way to define unemployment. When employed individuals are unable to find a suitable paid job for a predetermined period of time, they become unemployed. According to the Bureau of Statistics, as of September 2016, 38% of people in working age and 65% of young people in Nigeria were unemployed. Aguené (1991) defined unemployment as the number of people in the population who wish to work but are unable to do so because there are no jobs available.

2.2 Theories of unemployment

An essential idea in economics, the theory of unemployment rate examines the connection between the availability of labor (jobs) and the demand for labor (workers) in a particular economy. The percentage of the labor force that is actively looking for work but is having difficulty finding acceptable positions is referred to as the unemployment rate.

2.2.1 Classical Theory of unemployment

The two arms of the classical theory of unemployment, labour supply and labour demand, describe the factors that contribute to unemployment. Because labour demand is a derived demand that is negatively sloped downward, it follows that as wages rise, so does the demand for labour, and vice versa. Conversely, the labour force and the willingness to work or not (leisure), which is also a function of real wage, determine the labour supply. A distinct equilibrium of wages and employment is produced when labour supply and demand are in balance. The traditional theory of unemployment holds that there is never involuntary unemployment since the economy is always at full employment. This theory is based on the ideas of perfect competition and a free market, among other things. According to classical economics, unemployment may only arise when there are free market distortions, such as mismatches in the labor market brought on by an excess supply of labor compared to the demand for labor. This issue can be self-corrected by the market's invisible hand of labor supply and demand.

2.2.2 Keynesian Theory of unemployment

The British economist John Maynard Keynes' work "The General Theory of Unemployment, Interest and Money" is the source of the Keynesian theory of unemployment. The Keynesian Theory of Unemployment states that, under the assumption of fixed pricing, employment is connected with the effective demand for the goods and services that companies provide. Therefore, new jobs are created when collective demand produces an income at a certain price. This implies that employment changes in tandem with changes in overall demand. In terms of supply, Keynes suggested that although labor supply is fixed due to labor bargaining power, it still depends on real wage, as per classical theory. This implies that wages may remain constant even in the case of an excess of labor. A mismatch between the labor supply and the point at which spiral wages and labor demand converge leads to involuntary unemployment. In other words, cyclicalities or a lack of aggregate demand are the causes of unemployment, according to Keynesian theory. The two ideas outlined above have served as the foundation for a number of models that provide precise explanations for the prevalence of unemployment. These models include the search-matching model (Stigler, 1962; Phelps, 1970; McCall, 1970), the efficiency pay theory (Yellen, 1984), the competitive model of unemployment, worker motivation, and bargaining.

2.3 Empirical Literature Review

Numerous attempts have been undertaken to empirically investigate the factors that contribute to unemployment in various nations across the world, drawing inspiration from theories. There has been contention that a number of factors often influence unemployment. These variables can be internal or external, and they differ between nations depending on certain socioeconomic and political elements including institutions, national legal systems, and market conditions. First off, a thorough analysis of the impact of a few macroeconomic variables on Pakistan's unemployment rate from 1976 to 2013 was conducted by Magbool et al. (2013). They discovered that macroeconomic factors like GDP, population, inflation, and foreign direct investment (FDI) were significant short- and long-term drivers of unemployment in Pakistan using the autoregressive distribution lag model. To be more precise, their research indicated that every one of the previously stated factors was correctly signed, meaning that they had an adverse effect on unemployment. Similar to their findings, Oniore et al. (2015) concluded that there were both short- and long-term relationships between the rate of GDP growth, the rate of inflation, the degree of openness, and private domestic investment, and unemployment in Nigeria. Despite evidence that the GDP growth rate and inflation had a mitigating influence on unemployment, the degree of openness and the amount of domestic private investment exacerbated the situation in Nigeria. O'Nwachukwu (2017) observed that, in contrast to Oniore et al. (2015), real GDP had no discernible impact on Nigeria's unemployment rate; instead, the country's population, government spending, and inflation rate were the only factors that did. Baah Boateng (2013) investigated the relationship between supply and demand factors and unemployment in Ghana. He found that high growth leads to weak employment and that, in some cases, gender and education had a significant impact on the unemployment rate. A priori, reservation wages were found to exacerbate unemployment. Condratov (2014) conducted a literature review and identified the main factors influencing youth unemployment in Romania, including the country's demographic structure, economic environment, labor laws, and educational system. Kyei and Gyekye (2014) investigated the factors influencing unemployment in the province of Limpopo in South Africa. Through their research, they were able to demonstrate that while race, postgraduate degrees, middle age, gender, and elementary and incomplete secondary education were significant predictors of unemployment in the province of South Africa, GDP, male, matriculation, and youth had no discernible impact on the issue. Rather than concentrating on research unique to a single nation, Folawewo and Adeboje

(2017) took a look at the macroeconomic factors that influence unemployment throughout the West African region. Their results showed that while GDP did have a slight negative impact on unemployment, it was not statistically significant. Additionally, it was discovered that the population, labour productivity, and inflation all had a rising impact on unemployment. They also provided evidence that the effects of external debt and FDI on unemployment were negligible. Owing to some recent events on the global stage, economists are starting to look into other variables that affect unemployment. Recently, there has been a lot of discussion on issues like immigration and trade because people from other countries are concerned about how these policies may affect their ability to find employment. Latif (2015) conducted a study on the impact of immigration on unemployment. Using panel data collected at the province level in Canada and a range of econometric approaches, including FMOLS, DOLS, and VECM, the researchers concluded that immigration increased unemployment in the near term but decreased it over time. He also demonstrated a unidirectional Granger-causality between unemployment and immigration, demonstrating a causal relationship. Fromentin (2017) investigated the connection between immigration and unemployment in France. Immigration has no discernible long-term effect on the general unemployment rate, according to Isiaka Akande Raifu's research. According to the results of the VECM estimating approach, immigration helped to lower unemployment rates. The impact of immigration on unemployment rates in OECD nations between 1984 and 2003 was examined by Jean and Miguel (2011). Their results showed that immigration has a short-term impact on native native unemployment rates. Numerous research have been conducted to investigate how international trade affects unemployment rates, but the results have been mixed. As demonstrated by the research of Felbermayr et al. (2009), Dutt, Mitra, and Ranjan (2009), Kim (2011), Loganathan et al. (2011), Hassan et al. (2012), and Anyanwu (2014), a number of studies have found a negative association between trade openness and unemployment. However, a number of other studies have suggested that trade openness may result in higher unemployment, including those by Davis (1998), Egger and Kreickemeier (2009), Helpman and Itskhoki (2010), and Nwaka et al. (2015). Some research, such as Sener (2001) and Moore and Ranjan (2005), disagree with the contracting parties cited above and claim that trade openness has no direct impact on unemployment rates. Kim (2011) pointed out that even in cases where a declining effect was apparent, the impact of trade openness on unemployment rates was impeded by the labor market's rigidity and current restrictions. Despite the wealth of studies on the relationship between trade

openness and unemployment, the idea that the current account balance affects the unemployment rate has received little attention. However, there are few published publications that have attempted to examine this problem. A theoretical link between unemployment and the current account is established by Eaton, Kortum, and Neiman (2013) and Mayer (2010). Two empirical studies (Hojjat, 2014 and Ozer and Yeldan, 2016) particularly address the relationship between unemployment and the current account in addition to the theoretical study previously described. While Hojjat (2014) investigated the relationship between current account balance and unemployment in the United States, Ozer and Yeldan (2016) investigated the relationship between current account deficits and unemployment in Turkey. Hojjat (2014) used both descriptive analysis and the Ordinary Least Squares estimation method to find a significant association between unemployment and the current account balance. Improving the current account balance seems to be associated with a decrease in the US unemployment rate (Hojjat, 2014 and Ozer and Yeldan, 2016). Similar to this, but using a different methodology, Ozer and Yeldan (2016) used the Vector Error Correction Model (VECM) in conjunction with Variance Decomposition and Impulse Response methods to identify a unidirectional causal relationship between current account deficits and unemployment.

2.2. Types of Unemployment

There four types of unemployment in economics, these types emphasises on situations that warrant a particular unemployment in a country

2.2.1. Frictional Unemployment

The term "frictional unemployment" describes the short-term joblessness that occurs when people change jobs or enter the labour for the first time. It happens as a result of normal and expected labour market reasons, such as people moving to a new location, changing occupations, or looking for better employment possibilities. Frictional unemployment can be attributed to multiple sources. First off, there may be brief spells of unemployment as a result of the time it takes for people to look for appropriate job openings, apply, attend interviews, and negotiate employment arrangements. Furthermore, there may be frictional unemployment during the transition period if people relocate to places with stronger job markets or industries that match their skill set.

2.2.2. Cyclical unemployment

Unemployment that results from changes in the business cycle is referred to as cyclical unemployment. It happens when there is a slowdown or contraction in the economy, which lowers demand for products and services and, as a result, lowers the need for labour. Businesses may reduce production during economic downturns or recessions, which could result in layoffs or lower hiring, which would increase the number of unemployed people.

Cycles in unemployment can be attributed to various reasons. Economic cycles, which are defined as intervals of growth and contraction, have an impact on investment, consumer spending, and overall economic activity. Consumer demand tends to decline during economic downturns, which lowers production levels and lowers the need for workers. Particularly vulnerable to changes in the economy are the manufacturing, construction, and retail industries.

During economic downturns, policymakers frequently enact countercyclical policies to reduce cyclical unemployment. These actions could take the form of monetary policy changes, fiscal stimulus packages, or government expenditure initiatives designed to increase demand, improve economic activity, and generate employment. Cycle unemployment usually declines when the economy enters a recovery phase and conditions improve, as hiring by firms increases and demand for labour rises.

2.2.3. Seasonal unemployment

The term "seasonal employment" describes temporary work that is only available during certain periods of the year due to cyclical, predictable shifts in the demand for products or services associated with particular seasons, holidays, or occasions. Usually, this kind of work is found in sectors of the economy where seasonal fluctuations in labor demand occur. In agriculture, for instance, workers are frequently hired during planting or harvest seasons, when there is a greater need for labor, and their job expires at the conclusion of the season. Similar to this, a large number of jobs in the tourist and hospitality industries are generated during periods of high travel demand, such the summer or the winter vacation, when there is a marked increase in demand for services like lodging, excursions, and entertainment. In order to satisfy the increased demand around holidays like Christmas, Thanksgiving, or Black Friday, when consumer spending peaks, retail enterprises also hire seasonal workers.

Seasonal unemployment can be caused by a number of circumstances. In agriculture, planting and harvesting cycles have a direct impact on the availability of work. Similar to this, the demand for workers in the tourism industry varies according to holidays and vacation periods. Weather can have an impact on construction activities, which can cause seasonal variations in employment. Although some businesses view seasonal unemployment as a normal occurrence, it can be difficult for employees who depend on steady income all year round. Implementing legislation to support impacted workers during off-season periods, diversifying local economies to lessen reliance on seasonal businesses, and offering training programs to workers to learn skills in various industries are some ways to solve seasonal unemployment.

2.2.4. Structural unemployment

When there is a discrepancy between the requirements of open positions and the abilities and credentials of the labor force, it is known as structural unemployment. Structural unemployment arises from enduring transformations within the economy, including technological advancements, alterations in consumer preferences, or modifications to industrial frameworks, in contrast to cyclical unemployment, which is induced by fluctuations in the business cycle.

2.3 Reason/Causes of unemployment in Nigeria

The following are few reasons why there is high rate of unemployment in Nigeria:

2.3.1 Abandoned Agricultural sector

According history, the majority of Nigerian jobs have come from the agricultural sector, notably for the countries over 60% of workers. But in an effort to find oil crude, the music began to sound like the oil industry, which offers very little job opportunities to those who don't fit in. Despite the rise and development of the oil industry, the rate of unemployment continued to rise at a geometric pace.

2.3.2. Poor educational planning

Inadequate planning for education leads to a mismatch between graduates' abilities and what employers need, which is one of the main causes of unemployment in Nigeria. The practical skills and abilities required for employment are frequently lacking in graduates from educational systems that are out of step with the changing demands of the economy and industries. Inadequate access to high-quality education exacerbates this problem by limiting chances for skill development and employment, especially in rural areas and among marginalized people.

2.3.3. Poor enabling environment

In Nigeria, a lackluster enabling environment makes unemployment worse by impeding investment, economic expansion, and job creation. Both domestic and foreign investors are discouraged by factors including poor infrastructure, excessive bureaucracy, corruption, and insecurity, which restricts business growth and employment prospects. Furthermore, erratic electrical supplies, a lack of access to financing, and inconsistent legislation impede the expansion of small and medium-sized businesses (SMEs), which are important employers. Moreover, many people in Nigeria work in the unorganized sector, which is characterized by low productivity and underemployment due to issues like unstable markets, hard credit availability, and a lack of legal protections. Nigeria struggles to create enough jobs to accommodate its expanding labor population in the absence of a supportive climate for business and entrepreneurship, which maintains high unemployment rates and impedes socioeconomic progress. To address these problems and create an environment that supports long-term economic growth and job creation, extensive changes are needed in the areas of infrastructure, governance, regulatory frameworks, and financial access.

2.3.4. Gap between government policies and implementation

Because of the unpredictability, inefficiency, and inconsistent business environment caused by the gap between government plans and their implementation, unemployment in Nigeria is greatly increased. The efficacy of many policies intended at fostering economic growth and job creation is undermined by their inadequate execution because of bureaucratic delays, corruption, and a lack of accountability. Furthermore, policy direction is frequently interrupted by frequent changes in government leadership, which increases uncertainty for investors and enterprises. Moreover, inadequate monitoring and evaluation mechanisms fail to ensure that policies are effectively implemented, leading to missed opportunities for job creation and economic development. As a result, businesses face challenges in planning and making long-term investments, hindering their ability to expand operations and create employment opportunities. To address this issue, Nigeria needs to strengthen institutions, improve governance, and enhance coordination between government agencies to ensure the effective implementation of policies aimed at fostering sustainable economic growth and reducing unemployment.

2.3.5. Lack of entrepreneurial skill

Due to the restricted chances for self-employment and job creation, a deficiency in entrepreneurial skills is a major contributor to the unemployment rate in Nigeria. Innovation, job creation, and economic growth are all significantly influenced by entrepreneurship. Unfortunately, a large number of Nigerians lack the abilities, know-how, and capital required to launch and maintain prosperous enterprises. People are ill-prepared to handle the difficulties of entrepreneurship because the educational system sometimes places a higher value on academic credentials than on real-world experience and entrepreneurial training. A difficult business climate, regulatory obstacles, insufficient infrastructure, restricted access to capital, and other factors discourage would-be business owners from launching projects. Lacking the entrepreneurial abilities to recognize and seize business possibilities, a large number of Nigerians continue to depend on conventional employment, which is frequently insufficient to accommodate the country's expanding workforce. Improving the business environment, granting access to capital and mentorship, and enhancing entrepreneurship education are crucial for giving Nigerians the tools they need to launch and expand profitable enterprises, which will lower unemployment and promote economic growth.

2.3.6 Lack of employable skills and experience among the youths

The unemployment rate in Nigeria is largely caused by the youth's lack of experience and employable skills. Many recent graduates lack the technical competence, real-world experience, and practical skills that employers look for when hiring new employees. The educational system in Nigeria frequently prioritizes theoretical knowledge above providing students with the skills necessary for the workforce, which causes a mismatch between graduates' credentials and employer demands. Youths' inability to obtain practical experience and skills is further impeded by their restricted access to internship and vocational training programs. It is therefore difficult for young job searchers to obtain employment because many employers consider them to be inexperienced and ill-prepared. In addition, the swift speed of technological progress necessitates ongoing upskilling and adaptation, adding to the burden on young people without access to high-quality programs for training and development. Investments in initiatives that close the skills gap between education and employment, such as apprenticeship programs and vocational education, are necessary to address this issue by supporting younger ones the tools they need in order to

succeed in the workforce, these programs will lower Nigeria's unemployment rates and empower them.

2.4 Consequences of Unemployment

The problem of unemployment is an issue faced by most countries around the world , especially the third countries. Below are the consequences and effect of unemployment in a country.

2.4.1. Contributes to low GDP

It is a well-known economic truth that the size of the labor force directly affects the country's GDP. In addition to directly contributing to the production of manufactured goods, services, and agricultural products, the work force also raises consumer purchasing power.

In turn, it promotes economic growth. As a result, unemployment reduces a country's potential to grow its GDP (Njoku and Ihugba, 2011).

2.4.2. Psychological effects

Young people with two or more years of school and college experience have goals and objectives. After years of hardship, many people hope to find fulfilling employment, provide for their basic needs (food, shelter, clothing, and medical care), advance to a comfortable and respectable existence, and finally enjoy life's little joys. The psychological anguish of having their ambitions crushed may and does leave children with severe scars that are extremely challenging to deal with at such a young age. This pain can occur week after week, month after month. A person's sense of self-worth may be impacted by this trauma, which may result in clinical depression.

2.4.3. Family support:

Families usually depend on their child finishing high school or college and finding work to support the entire family in a country like Nigeria, where social security is nonexistent and people retire before they are really worn out. If that doesn't work out, the financial consequences are unbearably dire.

2.4.4. Political instability:

An increase in unemployment exacerbates a community's dissatisfaction with the incumbent administration. In the end, this leads to unstable coalition building or frequent government transitions. It does not have the capacity to set off a series of political upheavals, nor does it support long-term, stable economic strategy. "Our government's resolve to provide jobs for unemployed

men determines not only our future economic stability but also the stability of our democratic institutions," as President Franklin D. Roosevelt once said.

2.4.5. contribution on health:

It is also easy to understand how, in addition to the issues already discussed, a lack of steady income could lead to inadequate nutrition and have a detrimental effect on the health of youth and their families.

2.4.6. Contributes to crime and violence:

As youth unemployment rose, the wealth disparity grew, causing social unrest that might affect the entire town, state, and country (e.g., Niger-Delta insurgency, insurgent killings in the North-East and North-Central areas).

2.4.7. Unproductive labour force:

The loss of output in the form of goods and services is the most direct effect of an unproductive labor force on the economy, according to Njoku and Ihugba (2011). in the event that income taxes are not collected and that those who are liable to indirect taxes forfeit their receipts. The vast majority of young people have the required credentials

People with entrepreneurial talents who have graduated from universities with impressive accomplishments are at a disadvantage in the Nigerian workplace. The labor force is no longer productive as a result of this regrettable circumstance.

3. METHODOLOGY

The thesis emphasis time series date. The data was checked for time series properties. Firstly, I carried out An ordinary least square test of the my data to check the relationship between the unemployment and other variables with the help of GRETL which is a software for econometric analysis and statistical modelling. I also experimented the OKUN'S LAW that established the negative trade off relation between GDP and Unemployment. The dependent variable is unemployment measured in percentage while the independent variables include GDP, FDI, ICT imports, Wages and salaries etc.

3.1 The OLS regression

In OLS regression, the least squares approach is rather simple. Examine a scatterplot where the data points create a linear trend. Using an OLS linear regression algorithm to generate a line of best fit is the most accurate way to depict the distribution of the data points in a single line. The OLS method's line fit will have the minimum sum of squared deviations between each data point and the line, given the least squares property.

3.1.1 OLS regression procedure

The OLS regression analysis approach fits a regression plane onto a "cloud" of data (Fox, 2015) that is believed to show a linear trend. Although it does not touch every point in the data cloud, the regression plane represents the partial correlations between each slope (i.e., each regression coefficient "b") and the outcome variable while holding the effects of the other variables constant (Fox, 2015). The process of calculating regression coefficients in ordinary least squares involves minimizing the sum of the squares of the differences between the values of the observed data and those projected onto the regression plane. A researcher must carefully verify the numerous data assumptions involved in OLS regression before beginning the analysis for a number of strong reasons (see below).

3.1.2 OLS regression assumptions

Linearity- The linearity assumption states that a model cannot be properly presented if the independent variables do not all share a linear relationship with the dependent variable (Fox, 2015). Furthermore, all independent variables—aside from binary variables—must have a linear relationship with the dependent variable (Fox, 2015). This assumption is important since a non-

linear model cannot explain the systematic pattern of the connection between the dependent and independent variables (Fox, 2015). Because the units (or levels) of the independent variable in OLS are inconsequential—that is, every unit change in the independent variables always results in the same change in the dependent variable—nonlinearity skews the interpretability of the least squares estimators (Weisburd & Britt, 2014). This assumption can be assessed in a number of ways, both statistically and visually. One visual method to assess the linearity assumption is to look at a scatterplot of the studentized residuals compared to the unstandardized predicted values. This scatterplot can be used to examine whether the independent components and the dependent variable as a whole have a linear relationship. Partial regression graphs should be made to better illustrate whether there is a linear relationship between each independent variable and the dependent variable. Since the partial regression graphs will not be very useful, binary variables can be excluded from this procedure. Incremental F-tests, also known as "lack-of-fit" tests, can be used to assess linearity statistically by identifying any independent variables in the model that are unable to be accurately represented in a linear fashion with the outcome variable. These evaluations can show whether the model's variables have a significant impact on a deviation from linearity in the overall model. When there are no significant deviations, it is possible to determine that the linearity assumption is met, which means that a linear model is in line with the data.

Independence - If the regression model's error terms are uncorrelated, or independent of one another, then OLS's independence assumption is satisfied (Fox, 2015). The way the data were gathered greatly contributed to this supposition. Therefore, there probably wouldn't be any link between the error terms if data were taken at random from a large population. Nonetheless, the Durbin-Watson statistic can be used to statistically evaluate the premise that there might be autocorrelation across error components if there is cause to think so (for example, multilevel data). A Durbin-Watson test value of roughly two, with a probable range of zero to four, indicates weak correlation between residuals and suggests that the model's error components are independent (Rutledge & Barros, 2002).

- ❖ **Normality** - The normality assumption states that the distribution of errors, or residuals, must be uniformly distributed across the multiple regression plane (Fox, 2015). Fox (2015) outlines three reasons why checking for this assumption is essential, even when dealing with large datasets. First, the efficiency of OLS estimators is decreased when outlying data

points result in non-normal error distributions because the error distribution has "heavy tails" (Fox, 2015). Second, skewed error distributions can make it difficult to evaluate the least squares fit (Fox, 2015). This is because, given the predictors, the dependent variable's conditional mean is impacted by the skewed distribution (Fox, 2015). The best visual aids for assessing normalcy are quantile-comparison plots and kernel-density calculations. These graphical tests verify for normality by plotting studentized residuals onto a normal curve (kernel-density plot) and a line that represents the normal distribution quantile (quantile-comparison plot).

- ❖ **Multicollinearity** - Multicollinearity occurs when there is a nearly perfect linear relationship between two or more independent variables in an OLS multiple regression model (Fox, 2015). Multicollinearity can cause least squares estimator values to become unstable, meaning they could vary in response to even slight modifications in the data (Fox, 2015). Moreover, multicollinearity makes it difficult to identify the unique effects of each independent variable on the dependent variable (Weisburd & Britt, 2014). Two common techniques for assessing multicollinearity are to look at the bivariate correlations between each pair of independent variables and calculate the Variance Inflation Factors (VIF) of each predictor in the model. The first approach searches for correlations at or above .80, which is regarded as a strong indicator of multicollinearity in the model (Weisburd and Britt, 2014). The later method analyzes VIF values because they estimate the degree to which multicollinearity inflates regression coefficient variance (Fox, 2015). According to Fox (2015), VIFs should have a cutoff value of 10, with VIFs below 10 signifying the absence of multicollinearity in the model.
- ❖ **Constant Error Variance** - According to this assumption, the error variance, or the dependent variable's change around the regression plane, is constant (Fox, 2015). Regression models suffer from heteroskedasticity, or nonconstant error variance, which reduces the effectiveness of least squares estimators and can result in incorrect coefficient standard error estimations (Fox, 2015). A scatterplot of the studentized residuals plotted against the unstandardized projected values ($\hat{Y}$) can be used to evaluate this assumption. Whether the model has fulfilled this assumption can be determined by visually examining this plot. This assumption can be further evaluated statistically if heteroskedasticity is

assumed to exist and visual inspection does not yield unambiguous proof of homoscedasticity. Whether the model has fulfilled this assumption can be determined by visually examining this plot. This assumption can be further evaluated statistically if heteroskedasticity is assumed to exist and visual inspection does not yield unambiguous proof of homoscedasticity. A Breusch-Pagan test is one way to statistically evaluate this assumption. When the null hypothesis—that the model has homoscedasticity, or constant error variance—is tested, a significant result ($p < .05$) suggests that the dependent variable's fluctuation around the regression plane is heteroskedastic, or nonconstant. On the other hand, a nonsignificant p-value shows that the model has satisfied the homoscedasticity condition.

3.1.2 Interpreting OLS Regression results

When the OLS regression framework's assumptions are satisfied, a researcher can confidently interpret their findings. The researcher should look at roughly six statistics in a typical OLS linear regression output table: the p-value, 95% CI, unstandardized regression coefficient (b), standardized regression coefficient (β), standard error (SE), R^2 , and the F-statistic.

➤ P-value

The p-value in the regression output table shows the findings of hypothesis tests on the slopes of the independent variables.. The null hypothesis in these tests asserts that the independent variable's slope is equal to zero, meaning it has no bearing on the dependent variable's prediction. Regression coefficients are therefore significant predictors of the dependent variable if their p-values are less than .05.

➤ 95% Confidence interval

The 95% confidence interval for each unstandardized regression coefficient is shown in the majority of OLS regression output tables. The lower and upper bound values for the confidence interval are shown in this table column. The true value obtained in the population is 95% likely to fall between the lower and upper bound intervals.

➤ **Standardized Coefficients of Regression (β)**

Comparisons between independent variables in the same model are possible because to the standardized regression coefficient, also known as beta. This is because the model's variables—such as the number of previous arrests and treatment hours—are not quantified in the same units. Beta weights are used to compute standardized regression coefficients. Therefore, all unstandardized regression coefficients are entered into an equation and standardized based on the ratio of the standard deviation of the variable under study to the standard deviation of the dependent variable in order to determine the beta values for each variable (Weisburd & Britt, 2014).

➤ **Typical Error (SE)**

The regression output table's standard error illustrates the expected degree of error in the estimate relative to the population's actual value. Increasing the sample size lowers the estimate's standard error.

➤ **Unstandardized Regression Coefficients (b)**

In an OLS linear regression, the unstandardized regression coefficients display the slope of the line that connects the independent and dependent variables when the effects of the other variables in the model are held constant. Consequently, coefficients interpret that a one unit increase in X (the independent variable) corresponds to a predicted increase or decrease in Y (the dependent variable), independent of the influence of the other variables in the model. Because unstandardized regression coefficients, or insignificant slopes, are statistically similar to zero, it is important to keep in mind that only significant regression coefficients (i.e., when $p < .05$) should be evaluated. Unstandardized regression coefficients should be used to analyze findings from various studies.

➤ **R^2**

One can examine the coefficient of determination (R^2) to determine how well the whole model matches the data. This statistic shows the proportion of the dependent variable's variance that can be accounted for by the model's independent variables. In the social sciences, R^2 values of .40 or higher are generally regarded as strong. When analyzing the R^2 , it's important to keep in mind that the inclusion of more independent variables will simply result in an increase in the value. Instead, researchers can consult the modified version of R^2 . A measure of the percentage of variance in

the dependent variable explained by the independent variables, as opposed to the mean model (i.e., a model with no independent variables included using only the mean of the dependent variable), the adjusted R^2 is also adjusted for the number of predictors in the model (Fox, 2015). This adjusted statistic is based on the value expected to exist in the true population rather than the sample, as is the case with the R^2 (which results in a favorably biased estimation) (Fox, 2015).

➤ F-statistic

To ascertain whether the regression model is significant overall, the F-test statistic should be evaluated. The statistic shows if the addition of the independent variables makes the model significantly more accurate than the mean model at predicting the dependent variable. It is based on an omnibus test, such as the F-test, which is used in ANOVA. Additionally, a significant F-statistic indicates that the full model, which incorporates all independent variables, fits the data significantly better than the mean model. Furthermore, at least one of the model's regression slopes (b) deviates sufficiently from zero when the F value is significant.

3.2. The theoretical relationship between unemployment and GDP growth.

The checking and understanding the relationship between Unemployment and GDP growth is crucial in policy making, and theory of the economy. The relationship between the two is best described using the Okun's Law which displays an inverse relationship between unemployment and GDP growth as it states that for every 1% increase in unemployment rate, a country's GDP will result to additional 2% to 3% lower than its potential GDP

Okun's law can be expressed as followed:

$$\Delta U_t = \beta(\Delta Y_t - \gamma_t) + \epsilon_t + \alpha$$

Parameters

1. ΔU_t : Change in the unemployment rate at time t
2. α : Intercept term
3. β : Okun's coefficient, showing the responsiveness of unemployment to GDP growth
4. ΔY_t : Actual GDP growth rate at time t
5. γ_t : Potential or natural GDP growth rate at time t
6. ϵ_t : Error term

4.0 Results

This section talks about the study findings and result interpretations. This chapter provides a report using different analyses conducted on data. The challenges of data adequacy for this investigation are covered in the preliminary section. This chapter also covers the various method used for the analysis and interpretation of the results.

4.1 Ordinary least square regression

As we have a lot of variables that determine unemployment. I analyzed the relationship between unemployment and other variables such as FDI, Wages and salaries, ICT import etc using the GRETL software.

Model 1: OLS, using observations 2000-2022 (T = 21)

Missing or incomplete observations dropped: 2

Dependent variable: UNEMPLOYMENT

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	1.78901	0.798322	2.241	0.0489	**
GDPG	−0.0366584	0.0726016	0.5049	0.0246	**
ICT imports	−0.0270901	0.0276117	−0.9811	0.3497	
DCPS	1.16327	0.177517	6.553	<0.0001	***
WS	0.360787	0.0469315	7.688	<0.0001	***
Dcp of GDP	−1.15607	0.183460	−6.301	<0.0001	***
TFP	−4.78026	0.433907	−11.02	<0.0001	***
INFLATION	0.0232400	0.00908452	2.558	0.0285	**
Bank R E	−0.0304379	0.00759810	−4.006	0.0025	***
PLCS	0.969787	0.277520	3.494	0.0058	***
Mean dependent var	4.177286	S.D. dependent var		0.701187	
Sum squared resid	0.155949	S.E. of regression		0.124880	
R-squared	0.984141	Adjusted R-squared		0.968281	
F(10, 10)	62.05432	P-value(F)		1.20e-07	
Log-likelihood	21.68113	Akaike criterion		−21.36227	
Schwarz criterion	−9.872520	Hannan-Quinn		−18.86870	

4.1.1 The dependent variable

- Unemployment: This is the variable that the model is attempting to explain or forecast. Estimating the effects of various factors (the independent variables) on the unemployment rate is the goal of the model.

The model outputs:

- a. Coefficient: shows how each independent variable is thought to affect the dependent variable (Unemployment). While negative coefficients imply the contrary, positive coefficients indicate that an increase in the independent variable is linked to an increase in unemployment.
- b. Standard Error: evaluates the estimated coefficient's accuracy. More accurate estimates are shown by smaller values.
- c. P-Value: The likelihood that the coefficient deviates noticeably from zero. A smaller p-value indicates stronger evidence that the coefficient is not zero, meaning that it has a significant effect on the dependent variable. Coefficients with a p-value below 0.05 are regarded as statistically significant at the 5% level, which is a standard threshold.
- d. T-ratio: the coefficient's proportion to its standard error. It is employed to test the null hypothesis, which states that there is no effect and that the coefficient is zero. Stronger evidence against the null hypothesis is indicated by a greater t-ratio (in absolute terms).

4.1.2. Interpretation of each variable:

- ✓ Constants:

The coefficient is 1.78901. The unemployment rate is estimated to be around 1.79 when all independent variables are zero, yet depending on the situation, this may not be a significant number in reality. At the 5% level, the intercept appears to be statistically significant, as indicated by the p-value of 0.0489 (two stars).

- ✓ GDPG

The coefficient is -0.0367. this means there is a 0.0367 drop in unemployment for every unit rise in GDP growth. The p-value of 0.0246 (two stars) indicates statistical significance at the 5% level.

- ✓ ICT Imports:

This has -0.0271 coefficient. There is a 0.0271 drop in unemployment for every unit increase in ICT imports, although this effect is not statistically significant. The coefficient is not statistically significant, as indicated by the p-value of 0.3497.

✓ DCPS:

The coefficient is 1.16327. Unemployment rises by 1.16327 for every unit increase in DCPS. The effect is statistically significant at the 1% level, as indicated by the p-value of <0.0001 (three stars).

✓ FDI:

Coefficient: -0.1002. Although the effect is not statistically significant, there is a 0.1002 drop in unemployment for every unit increase in foreign direct investment.

✓ WS (Wage salary):

The coefficient is 0.360787. There is a 0.360787 rise in unemployment for every unit increase in wages and salary. At the 1% level, statistical significance is indicated by a p-value of less than 0.0001 (three stars).

✓ DCP:

The coefficient is -1.15607. This means that unemployment decreases by 1.15607 for every unit rise in DCP. At the 1% level, statistical significance is indicated by a p-value of less than 0.0001 (three stars).

✓ TFP (Total Factor Productivity):

The coefficient is -4.78026. Unemployment decreases by 4.78026 for every unit increase in total factor productivity. At the 1% level, statistical significance is indicated by a p-value of less than 0.0001 (three stars).

✓ INFLATION:

The coefficient for inflation is 0.0232400. This means that there is a 0.0232400 rise in unemployment for every unit increase in inflation. At the 5% level, the p-value of 0.0285 (two stars) denotes statistical significance.

✓ Bank R E:

The coefficient is -0.0304379. Unemployment decreases by 0.0304379 for every unit increase in Bank R E (or efficiency). At the 1% level, the p-value of 0.0025 (three stars) denotes statistical significance.

✓ PLCS:

The coefficient is 0.969787. Unemployment rises by 0.969787 for every unit increase in public legal and civil services. At the 1% level, the p-value of 0.0058 (three stars) denotes statistical significance.

Model fit:

- ❖ R-squared (0.9841): This shows that the model has a very high fit, explaining roughly 98.4% of the variation in the unemployment rate.
- ❖ Adjusted R-squared (0.96383): The number of variables in the model is taken into consideration by this modified form of R-squared. Even after adjusting for the number of predictors, it remains high, indicating that the model fits well.
- ❖ F-statistic (62.05432): The F-test determines whether there is a significant relationship between the dependent variable and at least one of the predictors. However, the whole model is statistically significant because to the extremely low p-value (1.20e-07).

Conclusion:

Variables like GDP growth, Dcps, Wage salary, Dcp, Tfp, Inflation, Bank R E, Plcs have a substantial impact on unemployment at various statistical significance thresholds. The majority of the variation in unemployment can be explained by the model, which has a strong fit (R-squared = 0.9841).

4.2 Okun's Law using Hodrick Prescott filter

The inverse link between changes in real GDP and changes in unemployment is explained by Okun's Law. It implies that unemployment tends to rise when the economy expands more slowly than its potential and to fall when the economy grows more quickly than its potential (or potential GDP). A common description of the relationship is:

$$\Delta U_t = \beta(\Delta Y_t - \gamma_t) + \epsilon_t + \alpha$$

Hodrick-Prescott (HP) filter: The HP filter aids in determining whether the variations in GDP and unemployment are structural (long-term trends) or cyclical (short-term swings). By assisting in the separation of the cyclical and trend components of economic variables such as GDP and unemployment, it can be crucial in the analysis of Okun's Law.

Fig1: Hodrick-Prescott (HP) filter diagram showing GDPG and the potential GDPG

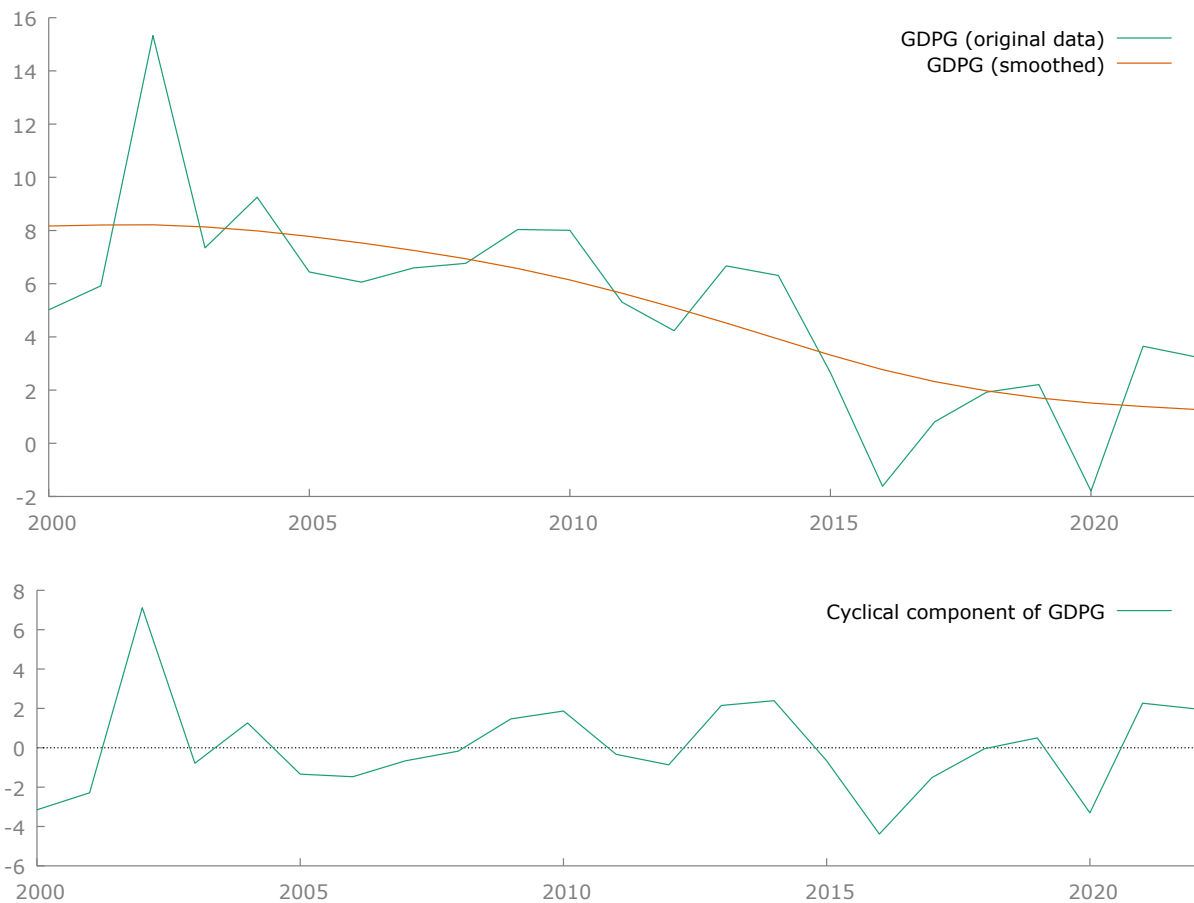


Figure 1

4.2.1 Explaining the graph

When $GDP < \text{potential GDP}$:

There is a negative gap which means that the economy is underperforming and there are unused capacity. Negative output corresponds with higher unemployment since the country is not utilizing all of its productive resources. The cyclical unemployment tends to be higher as firms are not producing as much as they should. This also result to slack economy, inflation and stagnation of a country.

When $GDP > \text{Potential GDP}$:

This is a situation where an economy is operating above its sustainable capacity and results to a positive gap. The economy is overheating, higher wage due to limited supply of labour, too low unemployment as firms will be hiring beyond the natural level of unemployment, unsustainable growth etc.

When $GDP = \text{Potential GDP}$:

There is a full employment and economy is at its Natural level of output with no inflation or deflationary pressures. Full employment, no output gap as there is no inflation, there is also a stable interest rate and monetary policy.

4.2.2 Effect of HP Filter on Okun's Law Analysis:

1. Improved Knowledge of Economic Cycles:

The HP filter aids in highlighting the economy's short-term dynamics by removing the long-term trend. This can be particularly helpful in comprehending how output and employment change during recessions and times of fast expansion. You may examine the relationship between the production gap—the cyclical deviation of GDP from potential GDP—and the unemployment gap—the cyclical deviation of unemployment from the natural rate—when you apply the HP filter to both GDP and unemployment.

2. Reduced Distortion from Long-Term Trends:

A greater understanding of the business cycle's impact on GDP growth and unemployment is made possible by the HP filter, which guarantees that the analysis is concentrated on short-term cyclical effects.

3. Cyclical Movement Sensitivity:

In particular, Okun's Law addresses short-term cyclical fluctuations in output and unemployment. Consequently, the filter's ability to accurately distinguish between trend and cycle is essential. The analysis's accuracy can be increased by using the HP filter to assist define these cyclical movements more precisely.

4.3 Findings

The percentage of the labor force, which consists of both employed and jobless individuals, that is not working but is actively looking for work is known as the unemployment rate. One of the elements of labor underutilization is unemployment. The 5.0% unemployment rate for Q3 2023 is a significant increase over the 4.2 rate recorded in Q2 2023. The sex-specific unemployment rates for men and women were 4.0% and 6.0%, respectively. In terms of region, the unemployment rate was 6.0% in cities and 4.0% in rural areas. The unemployment rate for young people was 8.6%. According to educational attainment, the unemployment rate was 2.7% for those without any formal education, 7.8% for those with post-secondary education, 6.3% for those with upper secondary education, 5.5% for those with lower secondary education, and 4.8% for those with primary education.

Figure 2: Analysis of Unemployment by Gender, Age, Educational Level, and Geographic Context

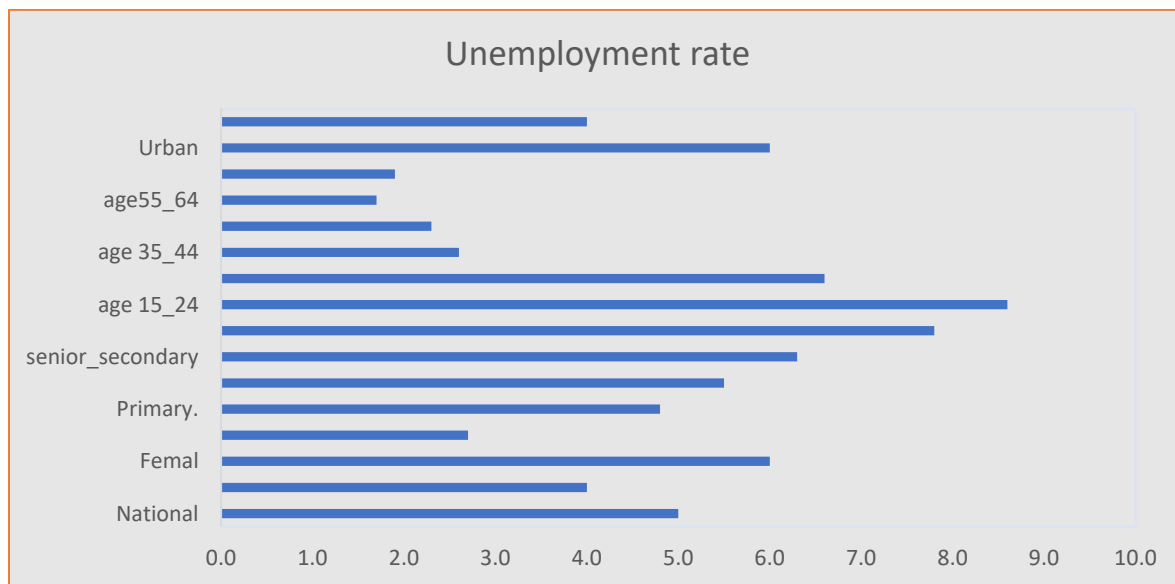


Figure 2

4.3.1. Effect of gross domestic product on unemployment in Nigeria

Like in many other nations, the relationship between GDP and unemployment in Nigeria can be examined through the prism of economic theory, especially Okun's Law, which postulates an inverse relationship between GDP growth and unemployment. Nonetheless, this relationship is impacted by some difficulties brought forth by Nigeria's unique economic circumstances. Growth in Nigeria's GDP frequently indicates that the country's economic sectors are increasing and may

be adding jobs. In theory, when economic activity rises, unemployment should decrease in response to a GDP gain.

4.3.2. Effect of credit to the private sector on unemployment in Nigeria

Since credit availability is a major factor in the expansion of the private sector, the creation of jobs, and the maintenance of economic stability, credit to the private sector may have a considerable impact on unemployment in Nigeria. The private sector is essential for absorbing labor, and credit availability fosters firm growth, innovation, and employee expansion. Credit to the private sector in Nigeria encourages enterprise by allowing firms to expand, invest in new technology, and boost productivity. Access to reasonable and sufficient finance enables businesses to boost output, which fuels economic expansion and increases demand for workers. Nigeria's economy is based primarily on small and medium-sized firms (SMEs), which also provide a substantial amount of employment. However, due to strict borrowing restrictions, high interest rates, and bank requests for collateral, SMEs frequently have trouble obtaining credit. Growth of loans to the private sector helps SMEs in particular, creating more job possibilities and lowering unemployment. Having more financing available will help businesses grow and create jobs, which will lower unemployment.

4.3.3. Effect ICT imports on unemployment in Nigeria

There are both direct and indirect effects of information and communication technology (ICT) imports on unemployment in Nigeria. Access to contemporary tools and technologies—which are essential for economic growth, productivity, and innovation—is made possible by ICT imports. If not properly handled, Nigeria's reliance on imported ICT products could pose problems as well. This is an examination, backed up by references, of how ICT imports impact unemployment in Nigeria. Nigerian companies may now obtain essential technologies like computers, software, telecommunications equipment, and internet infrastructure thanks to ICT imports. Businesses in a variety of industries, including banking, telecommunications, healthcare, education, and agriculture, benefit from these imports in terms of efficiency and productivity. Businesses that successfully incorporate these technologies are better able to grow their operations, which frequently results in an increase in job creation and a decrease in unemployment.

In order to grow their networks, telecommunications and internet service providers mainly rely on imported ICT equipment. For example, imports of servers, cell towers, and other communication

technologies have propelled the explosive growth of Nigeria's mobile telecommunications sector. This growth has directly decreased unemployment by creating opportunities for technicians, engineers, and customer service representatives. Nigeria's telecom industry's expansion is one such example. In the second quarter of 2021, the ICT sector made up around 17.92% of Nigeria's GDP, according to the National Bureau of Statistics (NBS). With direct and indirect jobs created in associated industries including telecommunications, software development, and IT support, the sector has been a major source of employment.

4.3.4. Effect of wages and salaries on unemployment in Nigeria

Minimum wage laws have been put in place in Nigeria to assist workers in meeting the country's minimum requirements. However, unemployment may result if the minimum wage is set excessively high in comparison to what companies can pay, particularly in industries that employ low-skilled workers. Because labor expenses are rising, employers may decide to hire fewer employees or even fire them. Nigeria has a high youth unemployment rate. If wages are set above the equilibrium level for low-skilled youth, companies might refrain from hiring, contributing to higher youth unemployment. Employers might not be able to flexibly alter their personnel if salaries are rigid—that is, insensitive to changes in the labor market or inflation. During recessions, firms may choose to fire employees rather than cut pay.

4.3.5. Effect of Gross Fixed Capital Formation (GFCF) on unemployment in Nigeria Gross Fixed Capital Formation (GFCF)

Gross Fixed Capital Formation (GFCF) has a crucial role in reducing unemployment in Nigeria by driving economic growth, enhancing productivity, and creating both direct and indirect employment opportunities. Investments in infrastructure, agriculture, and manufacturing can significantly lower unemployment, particularly if capital formation is directed toward labor-intensive sectors. However, challenges such as political instability, inadequate infrastructure, and capital-intensive investments in certain sectors may limit the potential impact of GFCF on unemployment. To fully harness the benefits of GFCF, Nigeria must address these challenges and encourage more investment in sectors with high job creation potential.

4.3.6. Effect of GDPG on unemployment in Nigeria Gross Fixed Capital Formation (GFCF)

Investments in buildings, machinery, and infrastructure (GFCF) increase productivity and promote long-term economic growth. More capital investment boosts industrial capacity and creates jobs,

especially in the manufacturing, services, and construction industries. In the near run, meanwhile, it might lead to the creation of more capital-intensive positions, which might not be sufficient to address unemployment, particularly if labor is replaced by labor-saving technology. GDPG and GFCF both have a good impact on employment, but in order to help Nigeria's workforce as a whole, they must be inclusive.

4.3.7. Effect of DCPS on unemployment in Nigeria

Direct Cash Payment Systems (DCPS), which are frequently observed in the form of cash transfers or subsidies, have the potential to impact Nigerian unemployment in a number of ways. The best way is by giving people financial support, DCPS can lessen immediate poverty, particularly in neighborhoods that are already at risk. This assistance might result in higher household expenditure, which would raise the demand for products and services. As a result, companies might grow and create jobs, particularly in the retail and agricultural industries. Furthermore, recipients of cash transfers frequently have the opportunity to invest in small firms, which promotes entrepreneurship and creates additional jobs. On the other hand, if DCPS is not properly planned, it may lead to reliance and discourage active job searching, especially if payments are not connected to employment or skill-development initiatives. Additionally, it can put a pressure on public coffers, preventing investments in vital infrastructure that might otherwise promote the long-term generation of jobs. DCPS has the potential to relieve short-term financial strains; nevertheless, its effect on joblessness is contingent upon the design of the program and its assimilation with wider economic strategies.

4.3.7. Effect of FDI on unemployment in Nigeria

In Nigeria, foreign direct investment (FDI) affects unemployment in both positive and negative ways. On the plus side, it increases economic diversification, enhances job creation, and advances skills, especially in industries like manufacturing and telecommunications. Additionally, it promotes the development of infrastructure, which creates more job opportunities. But foreign direct investment (FDI) can also result in job displacement, particularly if the foreign companies focus on capital-intensive industries like oil and gas or use expatriate labor. Additionally, as FDI concentrates in particular areas, regional imbalances may emerge. FDI generally lowers unemployment, although its effects vary depending on the kind of investment and the actions of the government.

4.4 Discussion

This study set out to evaluate the factors that contribute to Nigeria's high unemployment rate. The goal led to the creation of three objectives: developing and testing econometric models that can predict Nigerian unemployment trends empirically. Secondly, to list important macroeconomic factors that are highly correlated with unemployment in order to increase the prediction power of the models and, finally, to assess how well various economic forecasting techniques work and to determine which strategies work best in Nigeria.

Evaluating and comprehending the correlation between GDP growth and unemployment is crucial for formulating policies and understanding economic theory.

Conversely, exports of goods and services have a long-term, gender-neutral positive relationship with unemployment. Increased productivity and technological breakthroughs brought about by the arrival of businesses that attract foreign investment may be the main drivers of GDP growth. In response to this situation, production of products and services might rise more fast than demand. Nigeria is a country with abundant resources, a youthful and expanding population, and great economic and social potential. Unemployment is a significant problem there. Nigeria, with its riches of oil and human capital, has not been able to provide enough jobs to support its population. Being unemployed is a chronic issue that has shaped public opinion, influenced legislative choices, and fueled social discontent across time.

With a population of over 200 million, nearly half of whom are under 30, Nigeria is the most populous country in Africa. As a result, the labor force is constantly growing, putting more pressure on the economy of the nation to create jobs. High unemployment rates, especially among young people, are the consequence of the failure to create enough jobs to match the expanding work force. Nigeria's unemployment rate has changed over time as a result of a number of variables, including governance, security issues, international economic conditions, and economic policies. This essay addresses the unemployment rate in Nigeria and offers insights into its sources, patterns, and effects on the country's economy, society, and policymaking. The National Bureau of Statistics (NBS) reports that as of the first quarter of 2021, 33.3% of Nigerians were unemployed. This number is both among the highest in the globe and the highest in the history of the nation. Since the mid-2010s, the unemployment rate has been steadily rising. This era also happened to coincide with economic recessions and outside shocks that had a significant negative

impact on Nigeria's economy, such the decline in global oil prices. Nigeria has historically had a high unemployment rate. Even in times of economic expansion, it has been a persistent problem for many years. The problem's structural makeup indicates that it is linked to more serious problems with Nigeria's economy, including its dependence on oil, lack of economic diversification, inadequate infrastructure, subpar educational institutions, and corruption.

The string of recessions that Nigeria has recently seen has been one of the main causes of unemployment in the country. The severe drop in oil prices worldwide and the decrease in oil output brought on by terrorist activity in the Niger Delta region led to Nigeria's economic downturn in 2016. In Nigeria, oil constitutes more than 50% of government revenue and around 90% of export earnings. Consequently, the decline in oil prices had a catastrophic effect on the country's economy, resulting in job losses and a rise in the unemployment rate.

Nigeria's population is increasing at a pace of about 2.6% each year. Particularly among the youth, who make up over 60% of the population, the population is growing. Because millions of young Nigerians enter the work market every year, this demographic development puts enormous strain on the labor market. Nigeria's economy is not expanding quickly enough to accommodate the increasing number of job seekers, which presents a dilemma. High unemployment rates are a result of the large number of young people who continue to be unemployed or underemployed.

Nigeria's unemployment rate has also increased as a result of corruption and bad governance. Corrupt officials frequently misappropriate government funds that could be utilized to promote industrialization, infrastructural development, and educational reform in order to create jobs. This restricts the government's ability to successfully combat unemployment. Furthermore, because investors are frequently deterred by the absence of accountability and transparency in the Nigerian business climate, corruption impedes foreign direct investment (FDI). This further reduces the amount of room there is for private sector job growth.

Nigeria's over dependence on oil has hindered the development of other industries that have the potential to provide jobs. Although the oil industry gives the government a large amount of money, it is not a labor-intensive industry and only employs a small portion of the workforce. There is room for growth in other industries that have the potential to provide more jobs, like manufacturing. Nigeria's lack of industrialization and diversification has made it susceptible to outside shocks like changes in the price of oil. As demonstrated by the years after the oil price

crashes of 2014 and 2020, when oil prices decline, the economy shrinks and unemployment increases.

Many Nigerians are also stuck in low-paying, informal professions due to a lack of employment prospects, especially in rural areas. Because of this, social and economic inequality are increasing as the wealth gap between the rich and the poor keeps growing.

A large number of highly educated and talented individuals have left Nigeria in recent years in quest of better prospects outside, causing a major brain drain in the process. Doctors, engineers, IT specialists, and professors have left Nigeria in large numbers to travel to nations with better career prospects due to the country's lack of employment possibilities and unfavorable working circumstances.

Because it deprives Nigeria of the skilled labor force required to spur innovation and economic growth, this talent migration has detrimental effects on the nation's development.

5.0 CONCLUSION

To sum up, the unemployment rate in Nigeria continues to be a serious problem that presents significant social and economic difficulties. Nigeria has failed to provide enough job opportunities to support its expanding labor population, especially among the youth, despite having the greatest GDP in Africa. The nation's high rate of unemployment has been caused by a number of issues, such as a fast-growing population, a failing educational system, inadequate infrastructure, and an excessive reliance on the oil industry. Furthermore, exacerbating the problem are structural economic challenges, corruption, and political instability. The mismatch between graduates' skill sets and the demands of the job market is one of the biggest problems. Despite producing a large number of graduates each year, Nigeria's educational system does not give pupils the real-world skills necessary for gainful work. This is made worse by the nation's small industrial base, which is still underdeveloped in areas like manufacturing and other industries that typically employ significant numbers of people. Because of this, it is difficult for many Nigerians, especially those with higher education, to find work, which fuels social discontent and a growing tide of frustration. Despite good intentions, government initiatives to combat unemployment have yielded uneven outcomes. By offering chances for training and temporary employment, programs like the Graduate Internship Scheme (GIS), N-Power, and the National Directorate of Employment (NDE) have helped relieve some of the burden. However, because of operational difficulties, a lack of continuity, and insufficient money, these programs frequently fall short of providing long-term answers. Furthermore, a lot of these interventions are only meant to provide short-term respite; they don't deal with the root reasons of unemployment.

The private sector is essential to the job creation process, but it too faces a number of challenges that restrict its ability to hire workers. A lack of accessible finance, inadequate infrastructure, and an unwelcoming business environment impede the growth of Small and Medium Enterprises (SMEs), which are widely recognized as the foundation of employment generation in any given economy. A tiny portion of the workforce is employed by large, capital-intensive companies, particularly in industries like oil and gas. In the meantime, insecurity, corruption, and inconsistent policies prevent foreign direct investment—which may promote job creation—from reaching significant levels. With the rise of the gig economy and the ICT sector, in particular, technological developments provide some promise for lowering unemployment. Professionals with the necessary skills now have more options in industries including digital marketing, fintech, and

software development. Nevertheless, a sizable section of the populace, particularly in rural areas, is left out of the digital economy's benefits since it is restricted to those who have access to technology, education, and internet connectivity.

Moreover, agriculture continues to be underfunded and under-mechanized despite its enormous potential to create jobs. Insecurity, especially in areas impacted by banditry and insurgencies, lack of infrastructure, and difficult access to credit have hindered the government's attempts to revitalize agriculture through various programs. As a result, the industry can't take on a lot of young people without jobs.

In addition to economic factors, unemployment in Nigeria has significant social implications. The rising number of unemployed youths has been linked to increased crime rates, social unrest, and political instability. The frustration of being unable to find work leads many young Nigerians to engage in illegal activities, while others risk their lives attempting to migrate to Europe in search of better opportunities. This exodus of talented individuals not only represents a brain drain but also signals a deeper structural failure within the economy.

In the end, lowering Nigeria's unemployment rate would necessitate a complex strategy that takes into account both the supply and demand sides of the labor market. To guarantee that graduates have the skills required for the modern economy, the educational system needs to undergo considerable modifications on the supply side. Prioritizing vocational and technical training can help close the skills gap between job seekers and in-demand abilities. Furthermore, more entrepreneurial training programs must to be offered in order to promote independent work and the development of jobs, especially in industries like ICT and agriculture.

Demand-wise, the government needs to act more forcefully to foster an atmosphere that encourages corporate growth. This entails resolving the ongoing power supply problems that have impeded the expansion of the manufacturing sector, enhancing SMEs' access to reasonably priced finance, and maintaining a stable legislative framework that draws in foreign direct investment. To make business environments safer, security issues also need to be resolved, especially in rural areas where the concentration of employment-generating industries, such as agriculture, is found. In order to guarantee that marginalized groups—such as women and residents of rural areas—are integrated into the workforce, more inclusive economic policies are also required. In order to alleviate the persistent gender gaps in the workforce in Nigeria, specific policies need to be put in place. In a similar vein, as a sizable section of the population lives in rural areas, efforts should be

made to close the gap between them and urban areas by enhancing market accessibility and infrastructure. The unemployment problem cannot be solved just by the Nigerian government. To establish long-term job prospects, cooperation with the corporate sector, international organizations, and civil society is crucial. The possibility of addressing infrastructure deficiencies and promoting economic growth in important industries through public-private partnerships should be investigated. International agencies that support job development and poverty alleviation efforts include the World Bank and the International Labour Organization (ILO), which can also offer financial support and technical assistance.

Finally, the elimination of corruption is essential to any long-term solution to the unemployment issue in Nigeria. In Nigeria, corruption has long been a hindrance to efficient government, diverting funds that could be utilized to boost employment and the country's economy. To make sure that employment programs are not compromised by unethical behavior, it is imperative to fortify institutions, encourage transparency, and enforce responsibility.

In conclusion, the unemployment crisis in Nigeria is a complicated matter that calls for all parties involved to make thorough, persistent, and coordinated efforts. To address the underlying causes of unemployment, the government must implement both long-term structural reforms and immediate relief measures. By doing this, Nigeria would be able to fully utilize its people resources and achieve its economic potential, which will enhance living standards, promote economic stability, and pave the way for greater prosperity for all Nigerians in the future.

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